Sharp Convergence and Sample Complexity of Policy Mirror Descent for Average-Reward MDPs

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Motivation

Policy optimization (policy gradient (PG), TRPO, PPO) has a mature theory in discounted MDPs, but much less is known for **average reward**, a more natural objective when long-run performance matters, as in job scheduling and communication networks.

We develop a finite-time, finite-sample theory of policy mirror descent (PMD) directly for this setting.

Policy Mirror Descent

Ergodic MDP $(\mathcal{S}, \mathcal{A}, P, c)$: each policy π has a stationary distribution d^π, average cost $\rho(\pi)$ and differential action-value Q^π.
Goal: Find $\pi^\star \in \arg\min_\pi \rho(\pi)$, $\rho^\star := \rho(\pi^\star)$.
PMD update (actor) at every state s:

$$\pi_s^{t+1} \in \arg\min_{p \in \mathcal{P}(\mathcal{A})} \left\{ \eta_t \langle \tilde{Q}_s^t, p \rangle + D(p, \pi_s^t) \right\}$$

KL: natural policy gradient.
Squared Euclidean: projected Q-descent.
Critic $\tilde{Q}^t \approx Q^{\pi^t}$: tabular, $\tilde{Q}^t(s,a) = \hat{Q}^t(s,a)$; or linear function approximation (LFA), $\tilde{Q}^t(s,a) = \phi_{s,a}^\top w_t$.
Gap $\Delta_t := \rho(\pi^t) - \rho^\star$.

Technical Challenges

Discounting puts more weight on early costs, average reward puts equal weight on all costs:

$$\rho_\gamma = \lim_{H\to\infty} \sum_{k=0}^{H-1} \gamma^k c_k \qquad \text{vs.} \qquad \rho = \lim_{H\to\infty} \frac{1}{H} \sum_{k=0}^{H-1} c_k$$

Three challenges follow:

1. Distribution shift. Discounted analyses exploit the *initial* distribution to lower-bound state visitation. Equal weights in average reward make initial conditions transient, so we must control the *stationary* mismatch

$$\chi_t := \left\| \frac{d^{\pi^\star}}{d^{\pi^t}} \right\|_\infty \le \bar{\chi}$$

2. No effective horizon. Decaying weights give discounting a finite effective horizon $H_\gamma := 1/(1-\gamma)$. Equal weights give none: truncated value estimates are biased, and only mixing (t_{mix}) controls the bias.

3. Discounted surrogates break down. The common workaround analyzes ρ_γ and sends $\gamma \uparrow 1$. But discounted bounds scale polynomially in H_γ (up to H_γ^5 for policy gradient!), so the resulting average-reward bounds blow up.

Contributions

- **One master recursion** bounds the gap Δ_t for any critic; all our results follow from it.
- **Linear convergence** for exact, inexact-tabular and LFA updates, tight in the worst case. To our knowledge, the first non-tabular linear convergence to the optimum up to critic error.
- **Sharper sample complexity** than policy gradient: $|\mathcal{S}||\mathcal{A}|\, t_{\text{mix}}^3 \to |\mathcal{S}|\, t_{\text{mix}}^3 + |\mathcal{S}||\mathcal{A}|\, t_{\text{mix}}^2$ (tabular), $|\mathcal{S}||\mathcal{A}|\, t_{\text{mix}}^5 \to d\, t_{\text{mix}}^3$ (LFA).
- **Matching lower bounds:** both critics are optimal up to log factors.

Actor: Linear Rate

The critic's error $\tilde{Q}^t - Q^{\pi^t}$ enters only through its size along the PMD update and along the direction to $\pi^\star$:

$$A_t := \left| \mathbb{E}_{s\sim d^{\pi^{t+1}}} \langle \tilde{Q}_s^t - Q_s^{\pi^t},\, \pi_s^{t+1} - \pi_s^t \rangle \right| \le \bar{A}$$

$$B_t := \left| \mathbb{E}_{s\sim d^{\pi^\star}} \langle \tilde{Q}_s^t - Q_s^{\pi^t},\, \pi_s^t - \pi_s^\star \rangle \right| \le \bar{B}$$

Lemma 3.7 (Master recursion). If the step sizes grow geometrically, $\eta_t/\eta_{t-1} \ge \bar{\chi}/(\bar{\chi}-1)$, then for all $t \ge 0$,

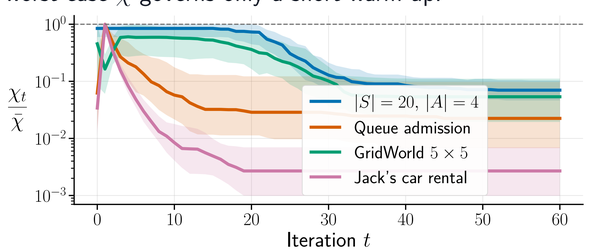
$$(\Delta_t - \underbrace{(\bar{\chi}\bar{A} + \bar{B})}_{\text{floor}})_+ \le \underbrace{\left(1 - \frac{1}{\bar{\chi}}\right)^t}_{\text{linear rate}} \cdot \underbrace{\Psi_0}_{\text{initialization}}$$

If $\eta_t/\eta_{t-1} \to \infty$, the rate improves to $1 - 1/(\chi+\zeta)$ for any $\zeta > 0$, with tail mismatch $\chi := \limsup_t \chi_t \le \bar{\chi}$.

▶ **Actor and critic separate:** the actor's machinery (step-size schedule, mismatch) sets the rate; the critic's error $(\bar{A}, \bar{B})$ affects only the floor.

▶ **No tuning needed:** doubling the step size every iteration works without knowing $\bar{\chi}$.

▶ $\chi \ll \bar{\chi}$ **in practice:** across four MDPs, $\chi_t/\bar{\chi}$ typically drops by 10–1000× within about 20–40 iterations, so the worst case $\bar{\chi}$ governs only a short warm-up.



Exact PMD

With the exact critic ($\bar{A} = \bar{B} = 0$):

▶ $\Delta_t \to 0$ and $\limsup_t \Delta_t^{1/t} \le 1 - 1/\bar{\chi}$.

▶ **Superlinear** when π^t converges (e.g., if the optimal policy is unique): $\chi = 1$, so $\limsup_t \Delta_t^{1/t} = 0$. KL and squared Euclidean geometries don't even need this!

▶ **Tight:** for every $\bar{\chi} > 1$ and $T \ge 1$, some MDP has $\Delta_t \ge \frac{1}{2}(1 - 1/\bar{\chi})^t \Delta_0$ for all $t \le T$, for any D and η_t.

Critics: Sample-Optimal

By Lemma 3.7, $\Delta_T \le \varepsilon$ follows from:

- **Critic:** $O(\varepsilon/\bar{\chi})$ accuracy, so the floor stays below ε
- **Actor:** $T = \tilde{O}(\bar{\chi})$ iterations, one critic call each.

Tabular critic: roll out once per state, not per action. In average reward, the Poisson equation gives

$$Q^\pi(s,a) = c(s,a) - \rho(\pi) + \underbrace{\sum_{s'} P(s' \mid s,a)}_{\text{one-step samples for every } (s,a)} \cdot \underbrace{h^\pi(s')}_{\text{rollouts once per state}}$$

LFA critic: average cheap pointwise estimates, not expensive critic estimates. A sublinear actor averages critic estimates ($\sim t_{\text{mix}}^3$ each) across its steps. Our critic instead takes K SGD steps on $\hat{Q}^t(s_k, a_k)$ ($\sim t_{\text{mix}}$ each) and averages the cheap iterates. At each actor step t:

$$w_{t,k} = w_{t,k-1} - \alpha\, (\phi_{s_k,a_k}^\top w_{t,k-1} - \underbrace{\hat{Q}^t(s_k,a_k)}_{\text{noisy label from one rollout}})\, \phi_{s_k,a_k}$$

$$w_t = \underbrace{\frac{1}{\lceil K/2 \rceil} \sum_{k > K/2} w_{t,k}}_{\text{averaging over SGD steps}}$$

Comparison

Rates. Prior guarantees are sublinear, or linear only with regularization, an exact critic, or a floor that does not vanish with the critic error.

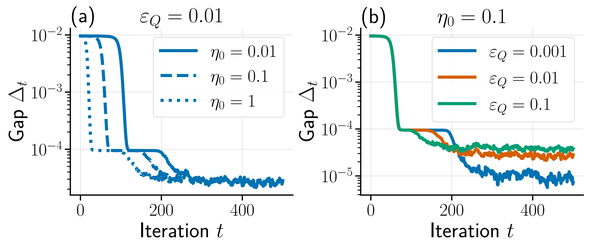
Samples to reach $\Delta_T \le \varepsilon$ (up to log factors, $\times 1/\varepsilon^2$):

	Tabular (generative model)	LFA (single trajectory)
Prior best PG	$\lvert\mathcal{S}\rvert\lvert\mathcal{A}\rvert\, t_{\text{mix}}^3$	$\lvert\mathcal{S}\rvert\lvert\mathcal{A}\rvert\, t_{\text{mix}}^5$
Ours	$(\lvert\mathcal{S}\rvert\, t_{\text{mix}}^3 + \lvert\mathcal{S}\rvert\lvert\mathcal{A}\rvert\, t_{\text{mix}}^2)\, \bar{\chi}^3$	$d\, t_{\text{mix}}^3\, \bar{\chi}^3$
Our critic LB	$\lvert\mathcal{S}\rvert\, t_{\text{mix}}^3 + \lvert\mathcal{S}\rvert\lvert\mathcal{A}\rvert\, t_{\text{mix}}^2$	$d\, t_{\text{mix}}^3$

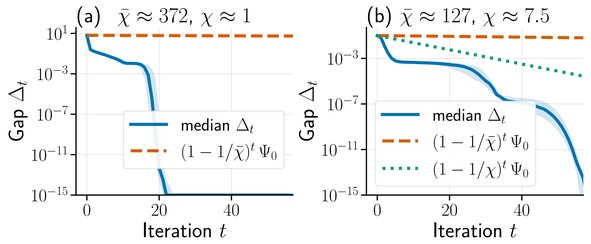
Prior bounds hold in expectation for a random iterate; ours hold with high probability for the last iterate.

Experiments

Step size sets the rate, critic error sets the floor.
We vary the initial step size η_0 and the critic noise ε_Q:
(a) Fixed ε_Q: larger η_0 reaches the same floor faster.
(b) Fixed η_0: the floor rises with ε_Q.



Convergence in practice is much faster than the worst-case rate suggests.
(a) Jack's car rental: $\chi \approx 1$, so the gap collapses superlinearly while the $\bar{\chi}$ curve barely moves.
(b) GridWorld: the tail mismatch is $\chi \ll \bar{\chi}$, and the gap falls below even the χ curve.



Takeaways

- **Average-reward PMD provably converges linearly.** The actor sets the rate; the critic's accuracy sets only the error floor.
- **The worst-case rate** $1 - 1/\bar{\chi}$ **is pessimistic.** In practice, it describes only a short warm-up, after which the tail mismatch $\chi \ll \bar{\chi}$ sets a faster rate.
- **Sample-optimal critics.** Rolling out once per state (tabular) and averaging cheap pointwise estimates (LFA) achieve sharper sample complexity than prior PG methods and match our critic lower bounds.

Project Page

Paper, code, slides and blog post:
enesarda.com/papers/average-reward-pmd

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